The Furpack - A Techpack

This document shows the design of the Furpack. A lightweight and secure backpack to transport the Furhat robot, its accessories and a laptop. In the development process the rough design and the main functionalities had been defined. In the first phase a 3D paper version showing basic functionalities and shape as well as a wearable backpack to show the use case were made. In the second phase two sewed backpacks were made – making the first usable prototypes with the rough design.

Basic stats

Load
Furhat robot (3,5kg; H410xW270xD240mm - check “ther infos” folder for more measurements); Accessories (0,5-2kg; cables and microphone); Laptop (±16inch; ±2kg) + charger; Total load (4-10kg)

Weight of the backpack
As light as possible (to make it comfortable) while providing enough protection for the expensive robot

Dimensions
As small as possible (**aim for carry-on luggage size H<550xW<400xD<250**), sewed prototype turned out H420xW300xD265 [mm]

Outside material
Should be water-resistant fabric

Furhat housing
Padded, thin, stiff walls like in camera bags (should withstand small drop heights)

Closing mechanism
Strong, long life, water resistant zipper for the four side pockets and the main opening

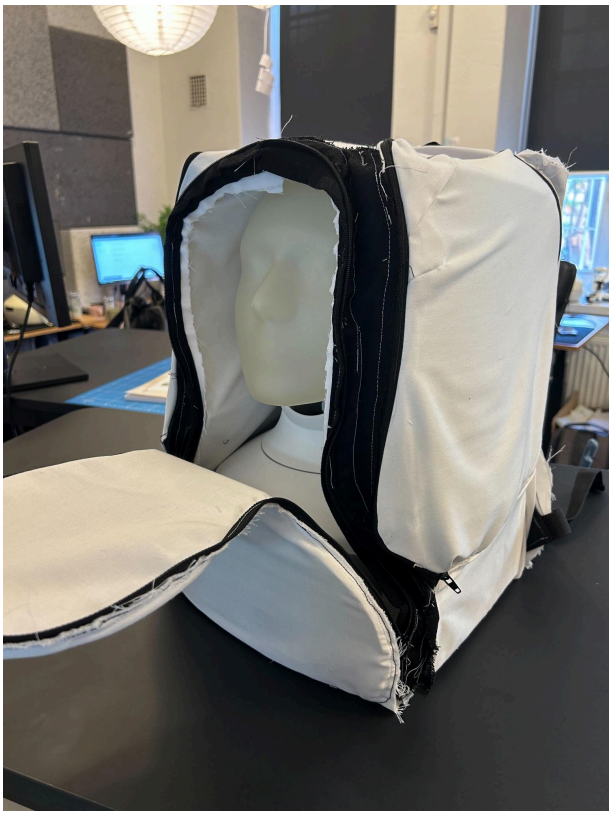
Wearing
Padded, comfortable, adjustable shoulder straps, maybe weight lifter straps

Extendability
Ideally the top lid is made out of a stretchable material or some kind of extendable mechanism so that the customer can carry more load if needed.

First sewed prototype

The sewn prototype is currently the best representation of the Furpak as we imagine it. It consists of curtain fabric, cardboard as padding, and simple zippers. The robot can already be completely stowed away and transported in it. Loading, unloading, and opening and closing the zippers works surprisingly well despite the unusual shape. At the same time, the zippers are not durable enough and the padding needs to be improved. The weight is well distributed with a medium load and the backpack is comfortable to carry. For larger loads, the carrying comfort would certainly need to be improved. In general, many functionalities and design elements are still missing. Many dimensions and shapes are very inaccurate due to production constraints and need to be adjusted. The **upper expandable pocket and the left secret pocket are missing** completely. The laptop pocket should also be closable with a zipper. The **second upper handle is also missing**. - Measurements: H420xW300xD265 [mm]. Please note that the measurements differ from the paper prototype as we were not able to keep the carry-on luggage limit with the padding we used.





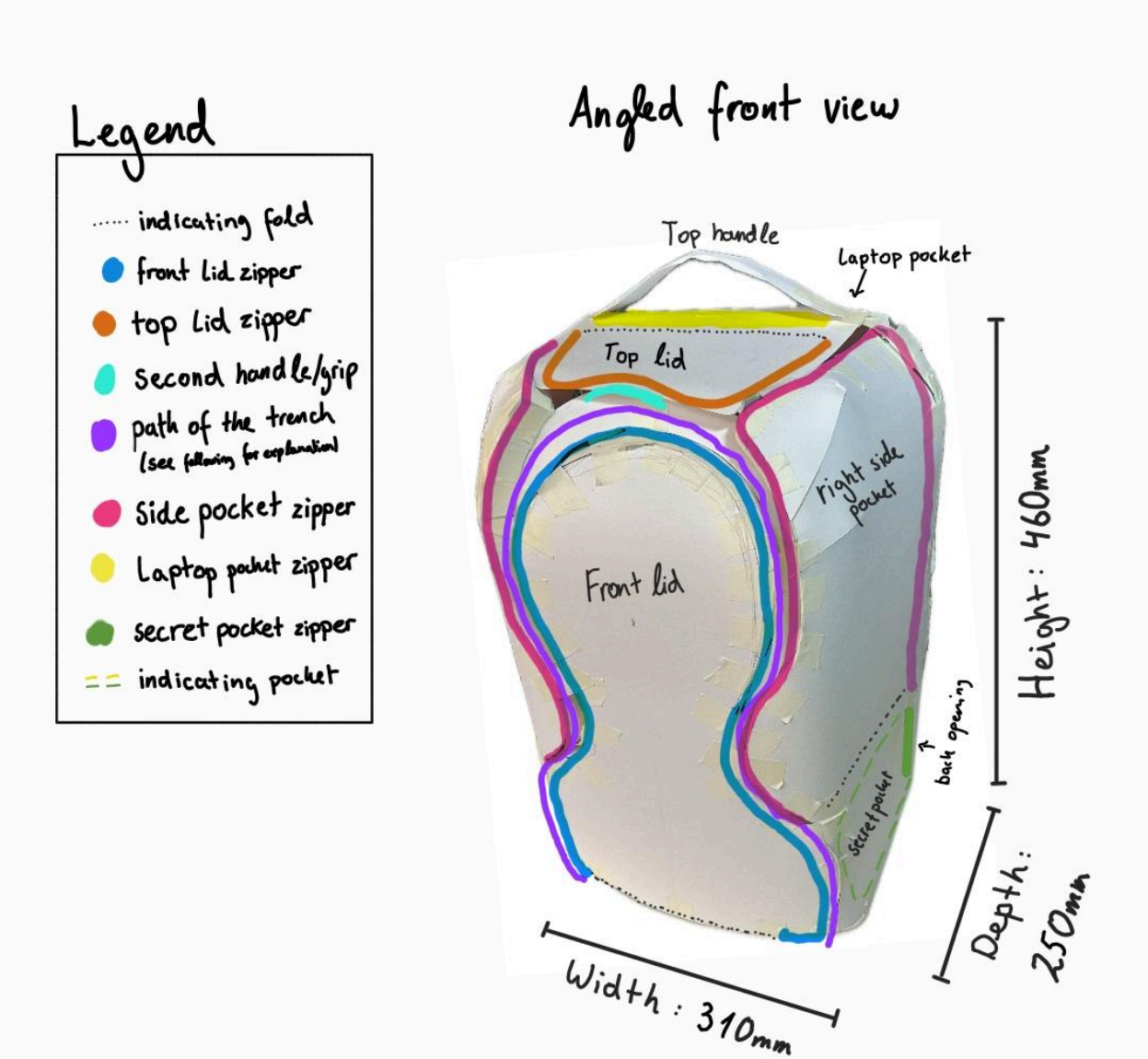






3D paper design

The 3D paper design was the first physical representation of the design. It was made solely from paper and tape. It shows the rough shape, the various lids, pockets, zippers, functions, and more. For illustrative purposes, these were marked and labeled in the following sketches. Please note that the measurements differ from the sewed prototype as we were not able to keep the carry-on luggage limit in the sewed prototype.

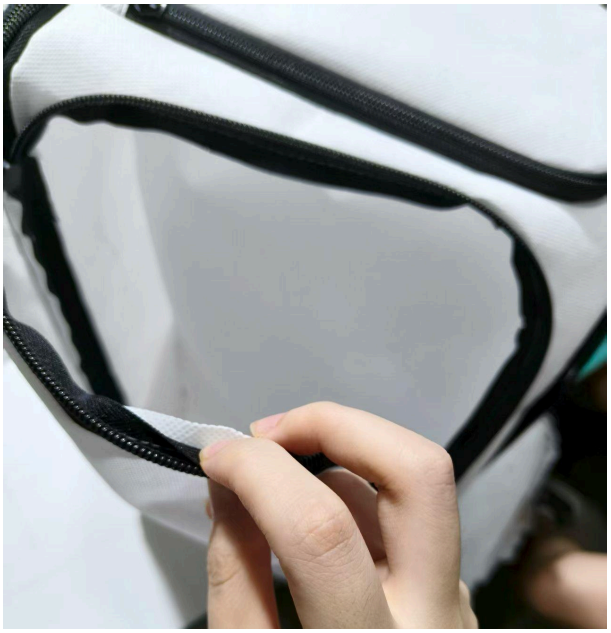


Second sewed prototype

The second sewn prototype was created externally and serves more as an initial design sketch. The structure and overall shape are already very good, but it is still too bulky and box-like. The large side pockets need to be better integrated into the shape of the front and the robot (see the colored markings). The sketch is also missing padding and structural reinforcement. Materials and colors have not yet been finalized.









Wearable version

An off the shelf backpack was modified for this version. The padded walls surrounding the Furhat, a neck support and padding in the lid matching the face and body contour were added. In the final design the contour should match the whole face and body and the padding should surround the robot smoothly, like a camera in a camera bag. The sewed prototype shows this concept very well but is lacking the fitted face and neck protection.



Possible colorways

The colorway is undecided yet. The shown versions are mainly to show the idea of the design trench and a rough direction the colorway could go. One option that has been discussed was adding small colorful highlights to make it more unique. There are also many ideas to make the design customisable for the customer. That will be discussed later in the project.

